Engineering - Undergraduate Level Assessment 1 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Find the solution for the equationX=AX+bu;X(0) = X_0 where A=|10|,b=|1|,X_0 =|- 1|, |- 22||2||0| = δ(t) (impulse function).

   (a) -3
   (b) 2
   (c) 0
   (d) 3
   (e) 10

2. Use partial fraction expansion to find the inverse z-transform of F (z) = [{2z^2 - 2z} / {(z - 3) (z - 5)^2}].

   (a) f [n] = 4nu[n] + [(5/6)n - 1] 6^n u[n]
   (b) f [n] = 6nu[n] + [(2/3)n - 1] 3^n u[n]
   (c) f [n] = 3^(n+1)u[n] - 5^n u[n]
   (d) f [n] = 3nu[n] + [(4/5)n - 1] 5^n u[n]
   (e) f [n] = [(2/3)n + 1] (3/2)^n u[n] - 2^(n-1) u[n]

3. A voltage source 60cos1000t V is in series with a 2-kΩ resistor and a 1 -. Findi_forced.

   (a) i_forced= 28cos1000t - 14 sin 1000tmA
   (b) i_forced= 24cos1000t - 12 sin 1000tmA
   (c) i_forced= 20cos1000t + 10 sin 1000tmA
   (d) i_forced= 30cos1000t - 15 sin 1000tmA
   (e) i_forced= 15cos1000t - 30 sin 1000tmA

4. The rating of a certain machine from the name-plate is 110 volts, 38.5 amperes, 5h.p. Find the input, output and efficiency at full load.

   (a) Input: 4235 watts, Output: 3600 watts, Efficiency: 85 percent
   (b) Input: 4500 watts, Output: 4000 watts, Efficiency: 88.9 percent
   (c) Input: 4350 watts, Output: 3730 watts, Efficiency: 85.7 percent
   (d) Input: 4100 watts, Output: 3450 watts, Efficiency: 84.1 percent
   (e) Input: 4200 watts, Output: 3900 watts, Efficiency: 86 percent

5. Find the Laplace transform of f(t) = t^2.

   (a) (2 / s^4)

(b) (1 / s^3)
(c) (t^2 / s)
(d) (t^2 / s^3)
(e) (2 / s^3)

True / False (10 marks)

Answer True or False

6. True or False: In a lap winding DC machine with 100 conductors and 10 parallel paths, the avera
7. True or False: A D-flip-flop is transparent when the output is independent of the input signal.
8. True or False: The function f(x, y) has a removable discontinuity at the origin (0, 0).
9. True or False: The power delivered by a roller chain of 3/4 in. pitch on a 34-tooth sprocket at 500
10. True or False: XOR and XNOR gates are universal gates that can implement any digital circuit by

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the process of converting temperature from Celsius to both the Fahrenheit and Kelvin scales, using -10C as a reference point. Include detailed calculations and implications in e
12. Derive the general solution of the differential equation dx/dt - 2(dx/dt) - 3x = 10asin(t) + b(2t + 1),

End of Paper